

Packing Circles in a Square: A Review and New Results

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1 Introduction

There are many interesting optimization problems associated with the packing and covering of objects in a closed volume or bounded surface. Typical examples arise in classical physics or chemistry with questions of the kind “what does the densest packing of atoms or molecules look like when a crystal or macro-molecule is formed with the lowest energy”. Also engineering and information science confront us with extremal problems associated with the packing and covering of objects. One of the most prominent examples arises from the study and design of spherical codes. A spherical code is a set of real vectors on the surface of the unit sphere in n -dimensional Euclidian space. In this case one searches for an arrangement such that the minimum separating angle between the vectors becomes as large as possible. One is interested in a solution for these problems, since spherical codes have important applications in the field of information processing. However, it was another closely related problem, the optimal packing of n equal circles in a square, which has fascinated mathematicians over the last few years. The circle packing problem is equivalent to the problem of scattering n points in a unit square, such that the minimum distance m between any two of them becomes as large as possible. The relation between the maximum radius r of the circles and the scattering distance m between the points is then given by $r = \frac{m}{2(m+1)}$. It is very surprising that such a problem, which at first looks rather simple, has brought to us a series of nice papers with a continuous improvement of the results. In this talk we give a short review on the history and the well known facts on the packing problem of n equal circles in a square, referencing recently obtained results. We present new results including the *proof of optimality* for up to $n = 20$ circles. And we calculate the exact expression for the optimal scattering distance, obtained within a *computer algebraic approach*. More precisely, we give the *minimal polynomial* corresponding to the optimal solution. In the following we discuss and summarize our findings.

$n = 2 - 9$: The situation for packing of up to 9 circles in a square was already solved in 1964. Here, the cases $n = 2, 3, 4$ and 5 are solved easily. For $n = 6$ it was R.L. Graham, mentioned in [1], who found the optimal solution. The proofs for $n = 7$ and $n = 8$ were done by J. Schaer [1], the one for $n = 9$ by J. Schaer and A. Meir [2]. Interesting about the optimal packing of 7 circles is the fact that one circle is “free”, i.e. its center can be moved within a bounded region.

$n = 10$: This problem has a long history which began in 1970 when M. Goldberg [3] proposed a symmetric arrangement consisting of 4 rows of 3-2-3-2 circles. The circles in this packing have a radius $r \sim 0.14706$. In 1971, J. Schaer [4] increased the radius to $r \sim 0.14777$, even though his packing contains two free circles in opposite corners of the square. Sixteen years later, R. Milano [8] proved that his packing with $r \sim 0.14792$ is the best *symmetric* one. But in 1989, G. Valette [10] found a better *chaotic* solution, which means that there is no symmetry in the arrangement of the circles in the square. The radius now was $r \sim 0.14818$. Also his solution did not survive very long. In the Abstract Index of the *Zbl. Math.* [11], we find the following note: “This packing has been improved, in 1989, by B. Grünbaum [12]; the radius of the circles is now approximately $r = 0.148197$.” We recently [14] found a packing with a radius $r \sim 0.148204$ and showed that it is the *optimum* one. When this work had been finished, we learned that this solution was already

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known 11 years ago and published by K. Schlüter [6]. It was also found very recently by M. Grannell [15], by J. Petris and N. Hungerbühler [16] and by M. Mollard and C. Payan [13]. None of them proved optimality, but Schlüter already conjectured that his solution may be the optimal one. Now we know that his conjecture was true. Furthermore, we are able to give the exact result for the optimal scattering distance. It can be obtained as the smallest positive zero of the polynomial

$$\begin{aligned} & 1180129m^{18} - 11436428m^{17} + 98015844m^{16} - 462103584m^{15} + 1145811528m^{14} \\ & - 1398966480m^{13} + 227573920m^{12} + 1526909568m^{11} - 1038261808m^{10} - 2960321792m^9 \\ & + 7803109440m^8 - 9722063488m^7 + 7918461504m^6 - 4564076288m^5 + 1899131648m^4 \\ & - 563649536m^3 + 114038784m^2 - 14172160m + 819200 \end{aligned} \quad (1)$$

$n = 11 - 13$: In [14] we have also presented the optimal solutions for 11, 12 and 13 circles. The proof of optimality was performed along the scheme presented in that paper using basically the same techniques as for $n = 10$. For 11 circles the optimal solution with a scattering diameter $m \sim 0.421280$ is only about 0.05% larger compared to the best previously known result of Goldberg [3]. The case $n = 12$ shows a simple diamond structure formed by 4 rows each of 3 circles leading to an m value of 0.388730. And indeed it is this highly symmetric structure which is the optimal one. Considering one more circle the situation becomes much more complicated. How to arrange an additional circle in the highly compact configuration of 12? There must be a complete rearrangement leading to a highly chaotic structure with an optimal scattering diameter $m \sim 0.366096$. The solution is difficult to find, since there exist two narrow gaps between two circles of the order 10^{-4} times the scattering distance. The optimal solution is more than 3.5% better than the regular 3-2-3-2-3 structure proposed by M. Goldberg [3] with $m \sim 0.35355$. Independent of us, M. Mollard and C. Payan [13] found the optimal packings for $n = 11, 13$ (and also for 14) without proving optimality. In all three cases we obtained the exact result as the smallest positive zero of a polynomial. The most difficult case was $n = 13$ resulting in a polynomial of degree 40.

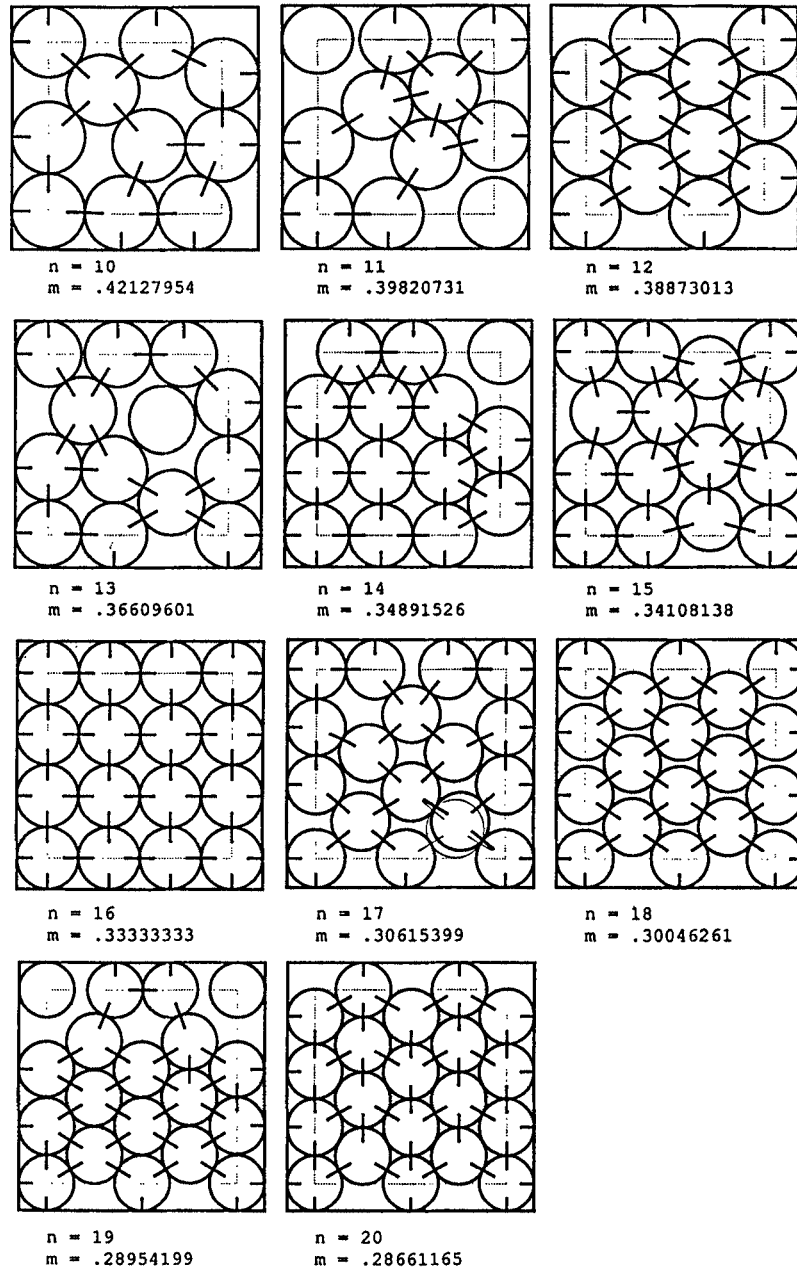
$n = 14 - 20$: In [14] we have also investigated *numerically* the problem for $n = 14$ to $n = 22$ circles. However, with an increasing number of circles, it becomes more and more difficult to converge to the global optimum. For even larger n it becomes more likely to get trapped in local optima containing hexagonal clusters. Nevertheless all solutions found were better or at least as good as those already known from the work of M. Goldberg [3]. One of the cases where we did not find the global optimum by optimization methods alone is for $n = 14$. J. Petris and N. Hungerbühler [17] showed that our packing can be improved by performing a few rearrangements. This motivated us to do some more work on our proof method which allows us now to present a uniform method for the range $n = 10, \dots, 20$. Fig. 1 shows the optimal packings for these values of n . Note that there are exactly two optimal packings for $n = 17$, one of which is axially symmetric. The two packings differ in the circle adjacent to the one in the lower right corner. For both positions of this circle, there is a free circle next to it. The packing for $n = 19$ is not symmetric. However, the lower 13 circles are invariant under a 180 degree rotation.

2 Notation and Definitions

Let us denote the closed unit square $[0, 1] \times [0, 1]$ by S . Then by $\mathcal{A}_{nm}$ we denote the *set of arrangements* of n points in S with minimal distance m , i.e.

$$\mathcal{A}_{nm} = \{ \{ \langle x_1, y_1 \rangle, \dots, \langle x_n, y_n \rangle \} \in S^n \mid (x_i - x_j)^2 + (y_i - y_j)^2 \geq m^2 \text{ for all } 1 \leq i < j \leq n \}$$

$\mathcal{A}_{nm}$ consists of one or more *connected components* which we call *packings of n circles of diameter m* . Hence, a packing P is a sequence $\langle P_1, \dots, P_n \rangle$, where P_i is the set of possible center points of the i -th circle. Two packings P and P' are considered identical if P' is obtained from P by a symmetry transformation and/or an index permutation. The i -th circle is called a *fixed circle* if P_i consists of a single point. It is called a *free circle* if P_i includes an open neighborhood.

Figure 1: Optimal packings for $n = 10, \dots, 20$

For a given number n , let m_n denote the maximal diameter $\max\{m \mid \mathcal{A}_{nm} \neq \emptyset\}$. As *optimal packings* we can then define the packings $P \subseteq \mathcal{A}_{nm_n}$. Given the number n , our problem consists now in finding m_n and a member of each optimal packing.

We will show that for $n = 17$ there are exactly two optimal packings, while for all other $n \leq 20$ there is only one optimal packing.

Successful termination (but not correctness!) of our method will rely on the assumption that circles in a packing are either fixed or free. This holds for optimal packings of $n \leq 20$ circles, but we don't have a proof for the general case. For small n , most of the circles in the optimal packings are fixed. Once the coordinates of the fixed circles are known, completion of the packing is obvious.

3 The proof method

For a fixed n , the following procedure has been used to find m_n and to prove its maximality.

Step 1: Find a good lower bound m for m_n . This is accomplished by Monte-Carlo optimization methods as described in section 3.1.

Step 2: For each optimal packing, find a small $2n$ -dimensional interval bounding the center points. Prove that every packing of n circles with a diameter $\geq m$ is contained in one of the intervals.

Step 3: Knowing the centers of the fixed circles up to a small tolerance, *guess* the optimal packing. That means: guess the connectivity graph and derive the diameter m_n from it. *Note:* A guess has to be made, because it is not clear after step 2, whether: (i) two circles seeming to touch each other indeed do, (ii) a circle seeming to have its center on the boundary of S indeed does.

Step 4: Prove that within its interval the guessed packing is *unique* for diameters $\geq m_n$.

Comment: Step 2 is successful only if step 1 produces a *good* lower bound. Step 4 is successful only if step 2 produces *small* intervals, if the set of fixed circles assumed in step 2 is correct, and if step 3 does the right guess.

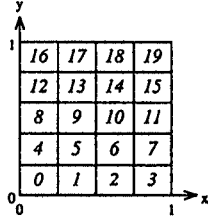
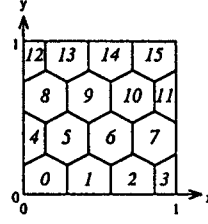
3.1 Finding sharp lower bounds

The search for the best radius was inspired by considering the problem as a typical numerical optimization task. Applying a standard BFGS quasi-Newton algorithm [9] and starting from a random initial position for the centers of the circles a local minimum of the objective function was found. Within about a dozen of trials we found for $n = 10$ in typically 30% of the solutions the optimal one. However, for larger number of circles the solutions got stuck in local, hexagonal substructures. Using a (stochastic) Langevin equation formalism, as described in [19] we were able to find near optimal or even optimal solutions.

3.2 The elimination procedure

We can now assume to have a lower bound m for the maximal diameter m_n . This rational number m is used to decompose S into a set of *tiles* in such a way that the *diameter* of any tile is $\leq m$. For complexity reasons, the number of tiles must be chosen small. Candidates for tile shapes are therefore *rectangles* and *hexagons*. As an example, Figs. 2 and 3 show good tilings for $n = 14$ and for the two kinds of shapes. (*Note:* m_{14} is greater than .348, while the tile diameters are approx. .320 and .345, respectively.)

Comment: Although using hexagons in general requires fewer tiles, we prefer rectangles, as we found that the elimination process converges faster when using them. The reason for this seems to be that rectangles are aligned in straight rows and columns.

Figure 2: Rectangle tiling for $n = 14$ Figure 3: Hexagon tiling for $n = 14$

If we denote by t the number of tiles, there are consequently n tiles which do contain a center point C_i and $t - n$ tiles which do not. We will refer to them as to “full tiles” and “empty tiles”. There are $\binom{t}{n}$ combinations for distributing $C_0, \dots, C_{n-1}$ among t tiles.

Note: Tiles are meant to be *closed*. Hence, points on tile boundaries set-theoretically belong to two (or more) tiles. However, we wish to assign each C_i to exactly one tile. So we have to accept that there may be more than one combination to which a single arrangement of circle centers can be assigned.

We will now fix our numbering conventions: *Tiles* are numbered as shown in Fig. 2. *Combinations* are first assigned a binary number, whose i^{th} bit is set iff the i^{th} tile is full. Then, combinations are ordered with increasing binary numbers, skipping those combinations which have a symmetric image with a smaller binary number.

For each combination, we will now try to show individually that it *cannot be assumed* by a packing $P \subseteq \mathcal{A}_{nm}$. This will succeed for most combinations (see Tab. 1).

We start with partitioning S into a grid of quadratic *cells*. It is advantageous to have cells contained in single tiles. We will call *active cells* those cells which are still candidates for containing a center point. Like tiles, cells are considered to be closed. Since we assumed a fixed combination, we can immediately reduce the set of active cells to those not completely covered by one or more empty tiles. *Note:* If a cell is removed (i.e. made inactive) next to an active cell, their common boundary is still present in an active cell.

As an example, Fig. 4. shows the active cells (unshaded) at this stage for $n = 10$, $m = .4212795$, a 4 by 4 tiling, a 32 by 32 grid and for the 500th combination.

Using the same example, Figs. 5 and 6 show what the elimination process does. One of the cells of tile 0 (shown at the left in Fig. 5) contains a C_i . Assume, this is the one drawn in black. Then the shaded area of tile 1 cannot contain a C_i . When repeating this for all active cells of tile 0, some cells of tile 1 lie each time in the shaded area. That means, these cells cannot contain a C_i and can therefore be made inactive, as shown in Fig. 6. This figure also shows the result of the next step, which is an even larger shaded area in tile 2 (on the right).

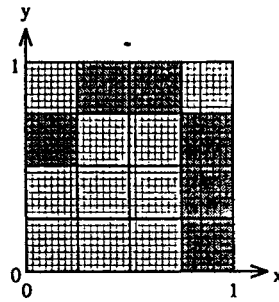


Figure 4: Active and inactive cells

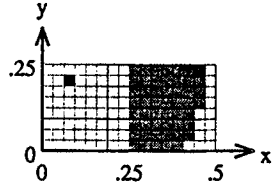


Figure 5: Tiles 0 and 1

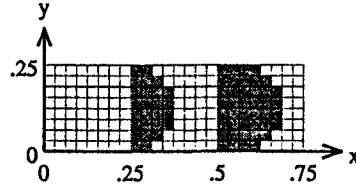


Figure 6: Tiles 0, 1 and 2

The procedure explained by this example is easily generalized into an algorithm working in both dimensions. It is iterated as long as cells can be removed.

If active cells remain, further progress can be achieved by refining the grid. Each cell is replaced by 4 smaller cells, which are active or not by inheritance. Of course the matrix is stored using run-length encoding, so computing time grows only with the number of active cells, not with the total number of cells.

For a few combinations however, the number of active cells grows fast enough to make further computation infeasible. To overcome this difficulty, two modifications of the algorithms can be made. Both are heuristics and in our implementation are called on user request.

Because free circles lead to large areas of active cells, they should be removed after a certain number of grid refinements. This is legitimate because to remove constraints for one of the centers means to generalize the problem.

The second kind of heuristics consists in splitting a problem (for a particular combination) into two sub-problems. This is done by selecting a coordinate variable x_i or y_i , ($1 \leq i \leq n$), and a threshold value k . Then the two cases $x_i \leq k$ and $x_i \geq k$ ($y_i \leq k$ and $y_i \geq k$, respectively) are treated separately. If k is well chosen, the number of active cells can be reduced more efficiently in the two sub-problems than in the original problem.

Tab. 1 summarizes the results of the unmodified algorithm, Tab. 2 those of the modified algorithm applied to the remaining combinations.

number of circles	lower bound m used (7 dec. digits)	tiles (rectang.)	finest grid used	initial number of combinations	remaining number of combinations	CPU seconds
10	.4212795	4x4	1024x1024	1051	1	17
11	.3982073	4x4	1024x1024	567	3	36
12	.3887301	4x4	1024x1024	267	1	13
13	.3660960	4x4	1024x1024	77	3	34
14	.3489152	4x5	1280x1280	9808	8	573
15	.3410813	4x5	1280x1280	3912	1	179
16	.3333333	4x5	1280x1280	1253	1	170
17	.3061539	5x5	1280x1280	136080	11	6567
18	.3004626	5x5	1280x1280	60645	2	8495
19	.2895419	5x5	1280x1280	22475	18	11551
20	.2866116	5x5	1280x1280	6814	2	11354

Table 1: Summary of the unmodified algorithm

number of circles	indices of optimal combinations	indices of other combinations	error radius	
			obtained	tolerated
10	951	-	.0011	.005
11	553	241, 275	.00005	.01
12	165	-	.00005	.01
13	55	54, 58	.0057	.01
14	1237	2050, 2054, 2510, 2528, 2951, 3223, 3605	.00004	.05
15	1367	-	.00004	.05
16	255	-	.00001	.1
17	40682 ³	40681, 42736, 46128, 46828, 55148, 114965, 114966, 114967, 115025, 115985,	.00009 .00032	.01
18	26230	56183	.00014	.01
19	12817	9721, 9799, 9800, 9801, 9802, 9803, 9804, 9805, 11210, 11273, 11274, 12885, 12992, 13002, 13007, 14540, 21793,	.0014	.0025
20	4942	4247	.00005	.02

Table 2: Indices of combinations which are hard to eliminate or contain optimal packings

3.3 Guessing optimal packings for $n = 10, \dots, 20$

The step treated in the previous section gave us for $n = 10, \dots, 20$ the packings shown in Fig. 1 up to a small tolerance which is listed in Tab. 2. Whenever two circles touch or one circle touches a square side this is indicated by a solid line in Fig. 1. Free circles are those without such lines.

It has to be verified that these arrangements represent packing as we defined them. First of all, algebraic *solvability* has to be checked. But we must also check that distances $\overline{C_i C_j}$ not declared to be m are greater than m and that the positions of any free circle center form a non-empty and connected set. For our range of n , these verifications are immediate.

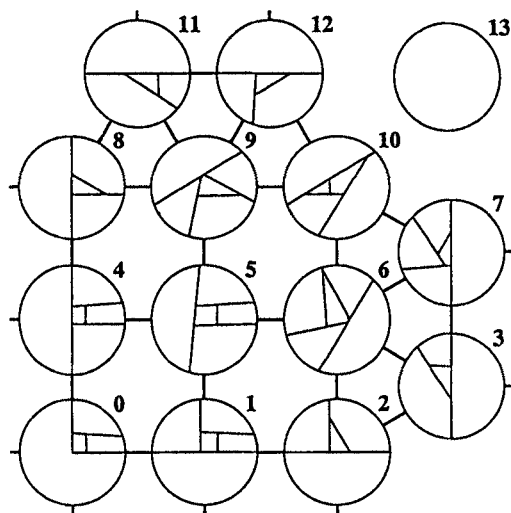
3.4 Proof of uniqueness

This step is somewhat similar to the elimination procedure that we already used. The main difference is that we don't work with sets of cells anymore but use convex regions bounded by lines and/or arcs. At the beginning these regions R_i are error circles around the C_i of the guessed packing. The radii r_i (see the right-most column in Tab. 2) are such that at least the $2n$ -dimensional interval obtained in step 2 is contained. If C_i lies on a square side, R_i is a half circle only (or a quarter circle if C_i is a corner of the square). The cutting process shown in Figs. 5 and 6 is now used in a modified form. The basic idea is again that a region R_i is being used to exclude parts of a second region R_j . Instead of cutting off a set of cells, we will remove everything lying outside a *straight line*. The endpoints of the line are found on the boundary of R_j by inspecting all "critical" points on the boundary of R_i . The goal is to reduce all regions belonging to fixed circles to *polygons* lying strictly inside the error circle. Fig. 7 shows the creation of such polygons for the values $n = 14, r_0 = \dots = r_{13} = .05$. For visibility, the regions have been magnified in this figure.

If this can be achieved, we have managed to reduce the error radii by a (common) factor $0 < q < 1$. The trick is now that, scaled down by q , the same sequence of cuts can be performed again. That means, we get sequences of concentric error circles which converge to the guessed optimal packing.

Fig. 8 shows why iteration is legitimate: Assume that the point B in the region R_j is determined by the point A in R_i . Then, the distances $\overline{C_i C_j}$ and $\overline{AB}$ are both equal to m . If A' and B' are chosen such that $\overline{C_i A'} = q\overline{C_i A}$ and $\overline{C_j B'} = q\overline{C_j B}$, then $\overline{A'B'} \leq m$. This is true even if the quadrilateral is concave or a "bowtie".

³This combination contains both optimal packings for $n = 7$

Figure 7: The uniqueness proof for $n = 14$

Again, this proof has been carried out on a computer. A strategy has been implemented to ensure that relatively wide segments are cut off at each step. In contrast to Fig. 7, for other values of n it may be necessary to use more than once each pair C_i and C_j of neighbors before a polygon is obtained. The highest number of cuts needed was 141, namely for $n = 13$.

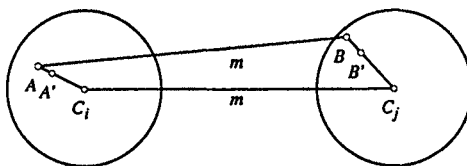


Figure 8: Reducing error radii

4 Finding the closed form solution for m

Given the optimal packings shown in Fig. 1 it is possible to try to solve for m exactly. The packings in Fig. 1 tell us which circles touch which other circles and which circles touch the boundary of the square. From this information, we can apply an elementary theorem of Pythagoras on right angle triangles to write down equations relating centers of circles to other circles. Adding boundary conditions, we obtain a system of polynomial equations to solve of degree at most 2. Then, in principle, it is matter of solving the equations. It is very difficult however to solve the larger systems of polynomial equations algebraically. So we try to simplify the system of equations if possible. In Fig. 1, the reader will notice that most of the packings have symmetries. From these symmetries we can derive other equations which will help enormously. Besides that, it turns out that there is a more effective approach to solving this particular problem based on the theory of *Gröbner bases*. Roughly speaking, given a system of polynomial equations, a Gröbner basis is an equivalent system in which

we hope to find that it has been triangularized. See Buchberger [5] for a reference on Gröbner bases. Since, in fact, we are only interested in finding the solution for one of the variables (in our case m) then all we need to do is make this variable least “last” variable in the Gröbner basis ordering. Consider the case $n = 10$ as shown in Fig. 1. Let the inner square have unit coordinates and m be the diameter of a circle. In the case $n = 10$ there is little symmetry. Indeed, it is one of the hardest cases to solve. However, there are four points forming a rhombus and three points lying on a straight line. Thus, for $n = 10$, we obtain the following system of 5 equations in 5 unknowns to solve

$$\begin{aligned} 1 - 2x_2 + x_2^2 + y_2^2 - m^2 &= 0 \\ x_2^2 - 2x_2m + y_0^2 &= 0 \\ 2 - 4x_6 + 4x_6^2 + y_6^2 + 2y_5m - 2y_5 - 2m &= 0 \\ 5 - 6m - 8x_6 + m^2 + 8mx_6 + 4x_6^2 + 4y_5^2 - 4y_5 - 4y_5y_0 - 4y_5m + 2y_0 + y_0^2 + 2y_0m &= 0 \\ 4x_6^2 + 1 - 2y_0 - 2m + y_0^2 + 2y_0m - 3m^2 &= 0 \end{aligned}$$

We compute a Gröbner basis (see Czapor [7] for a reference on the implementation in Maple [18] for the grobner package)

```
> grobner[gbasis]( [equations], [x[2],y[0],x[6],y[5],m], plex );
```

where the option `plex` means using a pure lexicographical ordering on the variables i.e. $x[2] > y[0] > x[6] > y[5] > m$. The Gröbner basis obtained contains the polynomial in m of degree 18 given previously. The complete Gröbner basis is too big to present here. It contains 9 polynomials whose coefficients are larger than 60 digits in length. The optimal solution for m then is the smallest real root of this polynomial. It would now be a simple matter to obtain complete solutions to our equations since the Gröbner basis is triangularized. We could now solve for m numerically and back substitute to obtain the solutions for the other variables. However, in this case, we can not solve the polynomial exactly in terms of radicals. The best we can do symbolically is to find this polynomial in m . We wish to mention that the `grobner` package in Maple contains a function `grobner[finduni]` that computes the polynomial in only m that we are interested in directly for us. I.e the best approach in Maple is to do

```
> grobner[finduni]( m, {equations} );
```

The minimal polynomials for the circles problems for $n = 10, 11, 13, 15, 17, 18, 19, 20$ are given below. The cases $n = 12, 14, 16$ are trivial. They are not included. The optimal solution for m is the smallest positive real root of these polynomials. We note that we were not able to find the solution for $n = 13$ using the Gröbner basis method. We found the solution using an elimination method based on computing polynomial resultants and greatest common divisors. Specifically, we made use of Maple's `resultant` and `gcd` functions.

For $n = 10$, $m \approx .421279$ from (1)

For $n = 11$, $m \approx .398297$ from $m^8 + 8m^7 - 22m^6 + 20m^5 + 18m^4 - 24m^3 - 24m^2 + 32m - 8$

For $n = 13$, $m \approx .366096$ from

$$\begin{aligned} &5322808420171924937409 m^{40} + 586773959338049886173232 m^{39} + \\ &13024448845332271203266928 m^{38} - 12988409567056909990170432 m^{37} + \end{aligned}$$

...

$$2960075719794736758784 m^2 - 174103532094609162240 m + 4756927106410086400$$

For $n = 15$, $m \approx .341081$ from $2m^4 - 4m^3 - 2m^2 + 4m - 1$

For $n = 17$, $m \approx .306154$ from $m^8 - 4m^7 + 6m^6 - 14m^5 + 22m^4 - 20m^3 + 36m^2 - 26m + 5$

For $n = 18$, $m \approx .300463$ from $13 - 144m^2$.

For $n = 19$, $m \approx .289542$ from

$$\begin{aligned} &242 m^{10} - 1430 m^9 - 8109 m^8 + 58704 m^7 - 78452 m^6 - \\ &2918 m^5 + 43315 m^4 + 39812 m^3 - 53516 m^2 + 20592 m - 2704 \end{aligned}$$

For $n = 20$, $m \approx .286612$ from $128m^2 - 96m + 17$.

5 Acknowledgment

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